

Foundations for nighttime lights data analysis

Ayush Patnaik^{1*}, Ajay Shah^{1*}, and Susan Thomas^{1*}

¹ XKDR Forum * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: 

Submitted: 12 September 2026

Published: unpublished

License

Authors of papers retain copyright
and release the work under a
Creative Commons Attribution 4.0
International License ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/)).

Summary

A remarkable development in alternative data is the use of satellites that measure nighttime light radiance. From the early 1990s, Nightlights (NL) has been used to measure economic activity. NL correlates well with standard measures of growth and development (Doll et al., 2006), which allows measures of economic growth where existing measures are either unreliable or lack the continuity and/or resolution of constant satellite observation.

Raw monthly NL files are not analysis-ready. They contain missing observations, negative values, outliers, background noise, and cloud-related attenuation (Jiang et al., 2017; Ma et al., 2014; Uprety et al., 2019). As a result, every research team must individually implement preprocessing, often with project-specific scripts. This creates entry barriers and weakens comparability and reproducibility across studies.

Statement of need

NighttimeLights.jl is a Julia package for cleaning and bias-correcting monthly nighttime lights from the Visible Infrared Imaging Radiometer Suite (VIIRS). It provides a transparent and reproducible workflow that implements established methods from the applied literature, including cloud-related attenuation correction (K Gupta et al., 2017; Patnaik et al., 2021; Pearson, 2002).

The package addresses a practical gap: many papers using NL data rely on individual preprocessing pipelines, making it difficult to reproduce findings or compare outcomes across studies. By consolidating key cleaning steps into one standard package, NighttimeLights.jl reduces duplicated effort and standardizes core data preparation.

State of the field

There are two dominant nighttime-lights products used in applied work: annual DMSP-OLS composites (1992-2013) and monthly VIIRS composites (2012-present). VIIRS has finer temporal and spatial resolution and is now the primary data source for contemporary analysis.

Most empirical NL workflows use combinations of missing-value recoding, outlier removal, and cloud-aware filtering, but implementation details vary across projects. This heterogeneity limits comparability. NighttimeLights.jl standardizes these operations while retaining modular control for users.

Several software tools already serve the nighttime-lights community, but they address a different part of the workflow. BlackMarbleR (Marty & Stefanini Vicente, 2024) and its Python counterpart BlackMarblePy (Stefanini Vicente et al., 2023) download NASA's Black Marble product and compute zonal statistics for administrative regions; Nightlights (Njuguna, 2019) does the same for the NOAA DMSP-OLS and VIIRS composites, though is currently unmaintained; and the VIIRS collections in Google Earth Engine (Gorelick et al., 2017) provide

server-side access and aggregation. General-purpose raster frameworks such as GeoWombat (Graesser et al., 2026) in Python and Rasters.jl in Julia supply the data-cube mechanics (reading, reprojection, alignment, and chunked computation) and, in the case of GeoWombat, radiometric corrections for daytime optical sensors, but they carry no nighttime-lights-specific cleaning. The nighttime-lights tools focus on acquisition and spatial aggregation, and the Black Marble product applies sensor, terrain, and atmospheric corrections upstream at the pixel level. None of them implement a configurable statistical cleaning pipeline for the raw NOAA monthly composites, and none correct the cloud-cover attenuation bias that dominates monthly series in monsoonal regions. Contributing these routines to the existing packages would mean maintaining the same methods across multiple language ecosystems while simultaneously overreaching their remit. Providing them as a standalone Julia package, with the cleaning pipeline itself as the unit of reuse, keeps the methods in one place and keeps a cleaned panel reproducible from a single package version.

Software design

NighttimeLights.jl is implemented in Julia and built on Rasters.jl for geospatial data handling. Data is represented as raster data cubes with dimensions (x, y, t) , and cleaning is organized as composable functions. Core functions include:

- Missing-data handling: `na_recode`, `na_interp_linear`
- Negative-value handling: `replace_negative`
- Outlier treatment: `outlier_hampel`, `outlier_variance`
- Background-noise filtering: `bgnoise_PSTT2021`
- Cloud-bias correction: `bias_PSTT2021`

These are wrapped into high-level pipelines such as `PSTT2021()`, `PSTT2021_conventional()`, and `clean_complete()`. This design supports both method-level customization and standardized, time-consistent cleaning pipelines.

Several design choices shaped the package. Primarily, cleaning is exposed as a set of small composable functions rather than a single opaque routine. This enlarges the API surface and requires the user to respect an ordering between steps, but it allows for customization and lets each stage be run, inspected, and switched out independently, which is what makes the package usable for methodological work while still allowing reproducibility. Secondly, the package is written in Julia because the core operations are per-pixel time-series computations over large pixel counts. Hampel filters, LOWESS fits, linear interpolation, etc. are awkward to express as fast vectorised array code and would otherwise require C extensions, if not for Julia's loop performance. Finally, existing best-practices cleaning and cloud-bias-corrected cleaning are kept as separately named and dated pipelines (`PSTT2021_conventional` and `PSTT2021`) so that results remain reproducible as well as directly comparable to the large body of prior work that does not correct for cloud cover.

Research impact statement

The package has been used as infrastructure for empirical work involving inference using VIIRS nighttime lights. Its main impact is methodological consistency: researchers can move from raw files to analysis-ready panels using explicit, shared preprocessing steps.

Because each stage is exposed as a public function, NighttimeLights.jl also serves as a platform for methodological development. Users can inspect intermediate outputs, compare alternative cleaning choices, and contribute new preprocessing methods while preserving reproducible workflows.

The cloud-bias correction implemented here was developed and validated in Patnaik et al. (2021), and NighttimeLights.jl is the reference implementation of that method. The package

has been used to build analysis-ready VIIRS panels for applied economic work, including a study of wartime economic activity in Ukraine (Hande et al., 2025) and an analysis of regional economic patterns across Java, Indonesia (Aviliani et al., 2026). It also underpins the public India Built and Lit dataset, which distributes cleaned nighttime-lights panels for India (XKDR Forum, 2026). Reproducibility is supported concretely: a Mumbai example cube ships with the package and is loaded with `load_example()`, every pipeline stage is a documented public function, the package is registered in the Julia General registry, and the results quoted in this paper were produced with the pinned versions listed under Computational details and can be regenerated with a linked Google colab notebook.

Computational details

Results referenced in this manuscript were generated with Julia 1.12.6 and NighttimeLights.jl 0.9.1. Julia is available at <https://julialang.org/> and the package is available at <https://github.com/xKDR/NighttimeLights.jl>.

Installation from the General registry:

```
] add NighttimeLights
```

Usage example

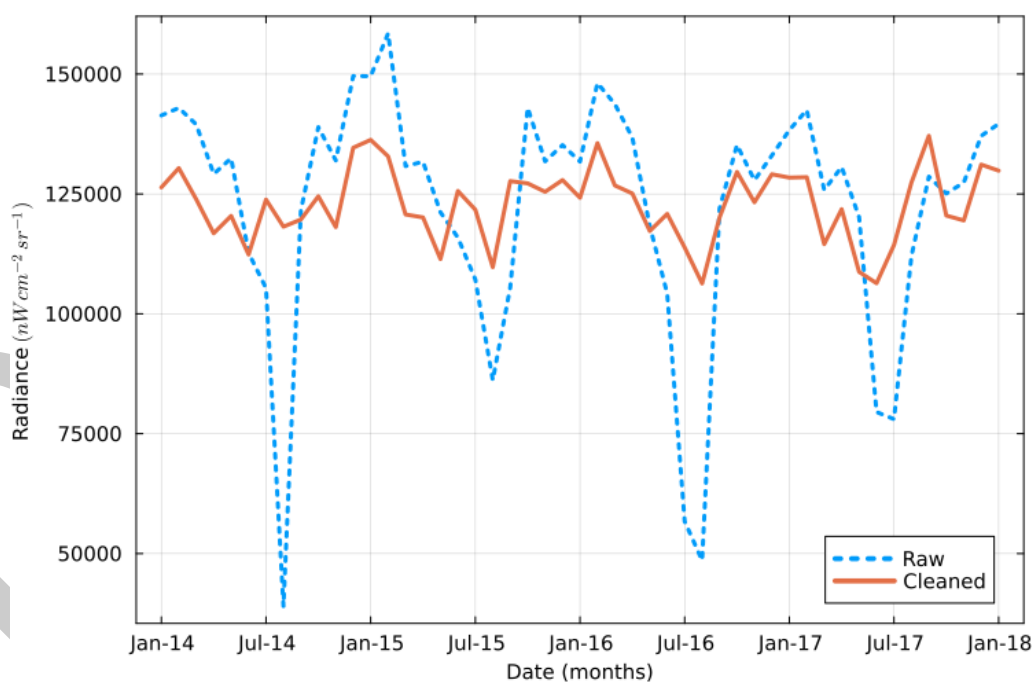


Figure 1: Aggregate radiance of raw and cleaned nighttime lights for Mumbai, January 2014 to January 2018.

Figure 1 shows the aggregate radiance of raw and cleaned NL of Mumbai. The notebook that produced this data can be found on Google colab [here](#). On the y axis, the lines show the aggregate raw and cleaned NL measured in $nW\ cm^{-2}\ sr^{-1}$. The x axis shows time measured in months from January 2014 to January 2018. Note the bias during the monsoon months (e.g. July), which is significantly reduced in the cleaned data.

Table 1: Effect of cleaning on nighttime lights for Mumbai.

Statistic	Raw	Cleaned
Minimum	-0.14	0.0
Mean	9.1	9.1
Median	3.2	0.0
99th percentile	54.0	59.0
Maximum	4300.0	1700.0

The table shows some summary statistics for raw and cleaned NL for Mumbai. Note the absence of negative values in the datacube. The maximum value after is also lower, as some outlier values have been removed. Half the values in the raw data cube are below 4, these can be attributed to background noise. All these measurements have been zeroed. The increase in the 99th percentile value is due to the bias correction procedure.

AI usage disclosure

No generative AI tools were used in the development of the software package, or the written text. AI assistance was used in formatting tables in the markdown document.

Acknowledgements

We acknowledge contributions from Anshul Tayal, Siddhant Chaudhary, and Hrishikesh Saikia to the codebase and package design. This work received financial support from the MIT Julia Lab.

References

- Aviliani, A., Kartawinata, M., Herawan, J. E., Parningotan, F., & Komala, A. (2026). When lights tell stories: The untold economic pattern of Java island. *Jurnal Ekonomi Indonesia*, 15(1), 1–18. <https://doi.org/10.52813/jei.v15i1.786>
- Doll, C. N., Muller, J.-P., & Morley, J. G. (2006). Mapping regional economic activity from night-time light satellite imagery. *Ecological Economics*, 57(1), 75–92.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18–27.
- Graesser, J., Mann, M. L., Hardtke, L., Denham, R., & Xu, S. (2026). GeoWombat: Scalable geospatial and remote sensing analysis in Python. *Journal of Open Source Software*, 11(125), 10812. <https://doi.org/10.21105/joss.10812>
- Hande, R., Patnaik, A., Shah, A., & Thomas, S. (2025). *Shedding light on the Russia-Ukraine war* (XKDR Working Paper No. 40). XKDR Forum. <https://www.xkdr.org/paper/shedding-light-on-the-russia-ukraine-war>
- Jiang, W., He, G., Long, T., & Liu, H. (2017). Ongoing conflict makes yemen dark: From the perspective of nighttime light. *Remote Sensing*, 9(8). <https://doi.org/10.3390/rs9080798>
- K Gupta, P., K. Srivastav, S., V. R. Sessa Sai, M., Gharai, B., Senthil Kumar, A., & V. N. Krishna Murthy, Y. (2017). Analysis of SNPP-VIIRS-DNB derived nightlights over India. *Asian Association of Remote Sensing*.

- 139 Ma, T., Zhou, C., Pei, T., Haynie, S., & Fan, J. (2014). Responses of suomi-NPP VIIRS-
140 derived nighttime lights to socioeconomic activity in china's cities. *Remote Sensing Letters*,
141 5(2), 165–174.
- 142 Marty, R., & Stefanini Vicente, G. (2024). *blackmarbler: Black Marble data and statistics*. R
143 package, Comprehensive R Archive Network.
- 144 Njuguna, C. (2019). *Rnightlights: Satellite nightlight data extraction*. R package version 0.2.4,
145 Comprehensive R Archive Network (archived).
- 146 Patnaik, A., Shah, A., Tayal, A., Thomas, S., & others. (2021). *But clouds got in my way:
147 Bias and bias correction of viirs nighttime lights data in the presence of clouds*.
- 148 Pearson, R. K. (2002). Outliers in process modeling and identification. *IEEE Transactions on
149 Control Systems Technology*, 10(1), 55–63.
- 150 Stefanini Vicente, G., Marty, R., Gadomski, P., SahitiSarva, & Garrett, R. (2023).
151 *BlackMarblePy: Georeferenced rasters and statistics of nighttime lights from NASA Black
152 Marble*. Python package. <https://doi.org/10.5281/zenodo.10667907>
- 153 Uprety, S., Cao, C., Gu, Y., Shao, X., Blonski, S., & Zhang, B. (2019). Calibration
154 improvements in s-NPP VIIRS DNB sensor data record using version 2 reprocessing.
155 *IEEE Transactions on Geoscience and Remote Sensing*, 57(12), 9602–9611.
- 156 XKDR Forum. (2026). *NighttimeLights and built-up volume dataset*. [https://xkdr.github.io/
157 India-Built-and-Lit/](https://xkdr.github.io/India-Built-and-Lit/).